Mandatory Covid-19 Vaccination Statuses of Residents across all Eight Regions of the United States of America: A Bayesian Random Effects Analysis of Selected Socioeconomic Factors

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1 Introduction

Following the COVID-19 pandemic and its many aggravating consequences spanning the incalculable devastation of lives and loss of jobs across all nations of the world[1], including the United States, a pivotal governmental concern to reduce the impact of this menace via a nationwide vaccination policy has been raised. This seemingly thought-through policy, notwithstanding, there seems to have been varying antithetical reactions across different people groups from the society against the government's policy.[2] The reason on the ground apparently supporting the pressure from the society to resist this vaccination policy is that it is being made mandatory, thus implicitly infringing on the fundamental human rights of the residents of the country. Some have even concluded the possibility of this policy being shrouded in

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political undertones, retorting that it may be to advance political agenda.[3] This research is geared towards investigating the selected factors that would or otherwise be the reason for the resistance of these pressure groups to the mandatory policy, and subsequently draw inferences that should or otherwise suggest a re-consideration of the policy.

The work considers a Bayesian mixed effects model[4] that analyses the effect of a selected combination of demographic factors of a two-level gender covariate and socioeconomic indicators of Education having four levels and Income having three levels on the mandatory Covid-19 vaccination status of a 1000 individuals randomly sampled from all fifty states in the U.S across eight geographical regions namely: South, West Coast, Mid-West, South-West, Rocky Mountains, New England, Mid-Atlantic and Great Plains, which were coded as 1 through 8 respectively.

2 Methodology

For a measure of whether or not an individual (sampling unit) takes the mandatory vaccine (at least one shot), let y be a binary response defined by the following piece-wise function as:

$$y = \begin{cases} 1, & \text{if a person took a shot,} \\ 0, & \text{otherwise.} \end{cases}$$

The logistic regression links y to $\boldsymbol{x}$ through: $\ln \frac{\pi}{1-\pi} = \alpha + \boldsymbol{x}'\beta$, where α is the regression intercept; β is the vector of coefficients given as $\beta = (\beta_1, \beta_2, ..., \beta_p)'$; and $\boldsymbol{x}$ is the vector of covariates stated as $\boldsymbol{x} = (x_1, x_2, ..., x_p)'$.[5] For the Bayesian analysis, the prior specifications of the parameters in generic form are such that $\alpha \sim \text{Cauchy}(\mu_\alpha, \sigma_\alpha)$; and $\beta \sim \text{Student_}t(\nu, \mu_\beta, \sigma_\beta)$, as suggested by Gelman (2008), where μ_α and σ_α represent the location and scale parameters of the cauchy distribution respectively; and ν, μ_β, and σ_β represent the degree(s) of freedom, the location and scale parameters of the Student_t distribution respectively.

2.1 Model Specifications

Given that $\boldsymbol{y} = (y_1, y_2, ..., y_n)'$ is the binary response n-vector for all persons sampled across all regions of the United States, $\boldsymbol{x}$ is a vector of socioeconomic predictors comprising Gender (made of two classes), Education level (made of four classes) and Income level (made of three classes).

For the purpose of this work, three different model specifications are considered in the ensuing sub-sections of this document. A first model is proposed that considers a homogeneous thread for all sampled persons across all regions. Second, we try to see how the analysis changes or otherwise by considering the model that accounts for varying covariate effects within regions but which stays constant across all regions. Lastly, we analyse the model in which covariate effects are allowed to stay constant within a region but made to vary for all regions.

For the Household Pulse Survey data[6] under consideration, the data is has 8 regional groupings with three unique number of predictors, namely, Gender, Education Level and Income Level. However, the categorical nature

of each of these three predictors is such that there are two classes for Gender, four classes for Education Level and three classes for Income level, making a total of 9 categorical dummies. In general, for any categorical variable with c classes, we undertake a dummy-coding of the $c-1$ sub-categories for each of the categorical variables. Specifically for the data under consideration, there are two classes for gender, three classes for Education gender subcategories amount to one, three and two for Gender, Educational Level and Income Level, respectively. Hence, this brings the total number of our covariates for model analysis to six. A break down of the indicator variables for each categorical dummy is as follows:

$$x_1 = \begin{cases} 1, & \text{if male} \\ 0, & \text{otherwise;} \end{cases}$$

$$x_2 = \begin{cases} 1, & \text{if Edu Level 1: less than high school} \\ 0, & \text{otherwise;} \end{cases}$$

$$x_3 = \begin{cases} 1, & \text{if Edu Level 2: high school grad} \\ 0, & \text{otherwise;} \end{cases}$$

$$x_4 = \begin{cases} 1, & \text{if Edu Level 3: more than high school and less than college} \\ 0, & \text{otherwise;} \end{cases}$$

$$x_5 = \begin{cases} 1, & \text{if Inc Level 1: less than } 25K \text{ dollars} \\ 0, & \text{otherwise;} \end{cases}$$

$$x_6 = \begin{cases} 1, & \text{if Inc Level 2: } 25K \text{ dollars and less than } 35K \text{ dollars} \\ 0, & \text{otherwise;} \end{cases}$$

Eventually, the vector of covariates becomes $\boldsymbol{x} = (x_1, x_2, ..., x_6)'$.

2.1.1 Model I: Homogeneity of Persons Across All Regions

This initial model assesses the probability that an individual takes the mandatory vaccine shot by considering homogeneity across all regions, while assuming away the regional effect on the model. Thus:

$$y_i \sim \text{Bernoulli}(\pi_i)$$

$\forall_i |\ i = 1, 2, ..., n$, resulting in the following estimation criterion:

$$\mathbb{P}(y_i = 1) = \pi_i = \frac{e^{\alpha + \boldsymbol{x}_i' \beta}}{1 + e^{\alpha + \boldsymbol{x}_i' \beta}}$$

where $\boldsymbol{x}_i = (x_{i1}, ..., x_{i6})'$; and α, β are constant parameters across regions.

2.1.2 Model II: Varying Regional Effect and Constant Covariate Effect Across All Regions

Under this scenario, we assess the existence of a varying intercept for each individual within a region but a fixed slope which stays constant across all regions. The baseline for a region is given as α_k. Thus, for the $K = 8$ regions:

$$y_{kj} \sim \text{Bernoulli}(\pi_{kj})$$

The model at region k becomes:

$$\mathbb{P}(y_{kj} = 1) = \pi_{kj} = \frac{e^{\alpha_k + \boldsymbol{x}_{kj}' \beta}}{1 + e^{\alpha_k + \boldsymbol{x}_{kj}' \beta}}$$

where: $k = 1, 2, ..., K$; $j = 1, 2, ..., n_k$; $\alpha = (\alpha_1, \alpha_2, ..., \alpha_k)'$; $\boldsymbol{x}_{kj} = (x_{kj1}, ..., x_{kj6})'$; and n_k = total number of observations within the k-th region. Thus α varies from region to region whereas β stays constant across all regions.

2.1.3 Model III: A Constant Regional Effect and Varying Covariate Effect within a Region

Unlike the preceding model, this scenario assesses the existence of a constant intercept for all persons within a region but a varying slope across all regions. The baseline for all regions is given as α. Similarly, for the $K = 8$ regions:

$$y_{kj} \sim \text{Bernoulli}(\pi_{kj})$$

The model at region k becomes:

$$\mathbb{P}(y_{kj} = 1) = \pi_{kj} = \frac{e^{\alpha + \boldsymbol{x}_{kj}' \beta_k}}{1 + e^{\alpha + \boldsymbol{x}_{kj}' \beta_k}}$$

where: $k = 1, 2, ..., K$; $j = 1, 2, ..., n_k$; $\beta_k = (\beta_{k1}, \beta_{k2}, ..., \beta_{k6})'$; $\boldsymbol{x}_{kj} = (x_{kj1}, ..., x_{kj6})'$; and n_k = total number of observations or units within the k-th region. Thus β varies from region to region whereas α stays constant across all regions.

For all the above $i = 1, 2, ..., n$, for $n = 1000$ observations and $\Sigma_{k=1}^{K} n_k = n$. Moreover, $k = 1, 2, ..., K$, for $K = 8$ regional groupings namely, South, West Coast, Mid-West, South-West, Rocky Mountains, New England, Mid-Atlantic and Great Plains, respectively.

Consequent to the above, we obtain the following likelihoods and prior specifications of the parameters as: $y_i \sim \text{Bernoulli}(\pi_i)$; where $\pi_i = \mathbb{P}(y_i = 1)$ is defined above based on the model under consideration; $\alpha \sim \text{Cauchy}(0, 10)$; and $\beta \overset{iid}{\sim} \text{Student_t}(1, 0, 0.03)$; following Gelman(2008).[7]

2.2 Estimation

On the basis of each of the foregoing, it is the case that, the Bayesian estimation of the posterior parameters should be obtained from the logistic regression where the linear predictor is linked to the log odds of taking the vaccine as previously shown. π indicates the probability that the observation took the mandatory shot. The logistic regression coefficient vector $\beta = (\beta_1, \beta_2, ..., \beta_p)'$ associated with a predictor vector $\boldsymbol{x}$ is the expected change in log odds of taking the mandatory vaccine shot per unit change in each element of $\boldsymbol{x}$. Hence, increasing the predictor by 1 unit multiplies the odds of taking the vaccine by e_j^{β}, for $\boldsymbol{x} = (x_1, x_2, .., x_p)'$ and $\boldsymbol{\beta} = (\beta_1, \beta_2, ..., \beta_p)'$. And the quantity e^{β} is considered the odds ratio associating $\boldsymbol{x}$ to y. For computational efficiency, an MCMC-based algorithmic Gibbs sampling procedure using STAN and python were used to obtain robust estimates for the posterior parameters.[8]

3 Data

The data used was from the Household Pulse Survey 2021 data taken from the Center for Disease Control(CDC)[6], whose original observation count stands at 47118 across several variables for various households of all fifty states of the U.S. However, for the purpose of computational efficiency, only a 1000 of these observations were randomly sampled across all states in order to undertake a very objective analysis of the underlying theory or concept.

4 Results and Analyses

The initial STAN outputs reveal estimates of the posterior means, standard errors of those means, and the standard deviations of the samples in the first three columns of the table, the middle five columns are quantile information of the estimates of the posterior parameters. R-hat values are metrics for measuring convergence. In effect, R-hat near 1 indicate approximate convergence, and R-hat greater than 1.1 suggests poor convergence. For convergence diagnostics in STAN, the rule of thumb is that R-hat be less than or equal to 1.1 for all parameters. Also, "lp" in STAN refers to the log likelihood of the observations conditional on the posterior parameters.[8]

For the first model which assumes only fixed effects for a pooled fit, the intercept value of -2.45 indicates that the probability of taking the mandatory vaccine in absence of any socioeconomic factor is $\left(\frac{e^{-2.45}}{1+e^{-2.45}}\right) = 0.08$ That shows a low probability of taking the vaccine. The odds of taking the mandatory vaccine as against not taking the same, by a male, drops by 4% compared to that by a female, when all other factors are held constant. On the basis of education levels, the odds of taking the mandatory vaccine versus not taking it increases by 2% and 134% respectively as a person moves down from the $4th$ education level (college grad) to the $1st$ (less than high school) and $3rd$ (more than high school and less than college) education levels, whereas it increases by 45%, without recourse to gender and income positions. As pertains to income levels, it can be observed that $1st$ (less than 25K dollars) and $2nd$ (greater than 25K but less than 35K dollars) level income earners are more likely to take the mandatory vaccine as the odds respectively increase by 11% and 30% as a person moves down from the $3rd$ income level (35K dollars and more), "all other things being equal." All the above results can be gleaned from the output in table 1 below. Also included are the convergence diagnostics trace plots and posterior distributions plots in figure 1. A plot of the change in odds for each of the predictor variables, while holding the remaining factors constant, is shown in fig 2. At a midline of zero (representing a 100% benchmark), it can be seen that the odds for taking the vaccine by a male gender reduces by approximately 4%. On the other hand, the odds for taking the vaccine by each of the remaining variables increases, while holding the remaining factors constant. In this regard, the highest increase occurs as a person moves down from Educ Level 4 to Educ

Level, with approximately 133%.

Table 1: Output for the Fixed Effects Model

```
Inference for Stan model: anon_model_07915228d8d4d4110c2b476607c54bb0.
4 chains, each with iter=1000; warmup=500; thin=1;
post-warmup draws per chain=500, total post-warmup draws=2000.
```

	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
alpha	-2.45	9.3e-3	0.18	-2.83	-2.56	-2.44	-2.32	-2.12	365	1.01
beta[1]	-0.04	5.3e-3	0.11	-0.34	-0.06	-0.01	10.0e-3	0.09	413	1.0
beta[2]	0.02	0.01	0.17	-0.17	-0.02	4.5e-3	0.04	0.39	224	1.02
beta[3]	0.37	0.03	0.43	-0.06	0.01	0.16	0.73	1.33	205	1.02
beta[4]	0.85	0.01	0.25	0.33	0.68	0.85	1.02	1.34	352	1.01
beta[5]	0.1	0.01	0.2	-0.08	-5.2e-3	0.02	0.12	0.72	188	1.01
beta[6]	0.26	0.02	0.29	-0.05	0.02	0.16	0.48	0.88	255	1.01
lpred[1]	-2.49	9.8e-3	0.19	-2.87	-2.62	-2.48	-2.36	-2.15	367	1.01
lpred[2]	-2.45	9.3e-3	0.18	-2.83	-2.56	-2.44	-2.32	-2.12	365	1.01
lpred[3]	-2.49	9.8e-3	0.19	-2.87	-2.62	-2.48	-2.36	-2.15	367	1.01
lpred[4]	-1.64	8.5e-3	0.2	-2.07	-1.77	-1.63	-1.5	-1.29	556	1.0
lpred[5]	-2.49	9.8e-3	0.19	-2.87	-2.62	-2.48	-2.36	-2.15	367	1.01
lpred[6]	-2.45	9.3e-3	0.18	-2.83	-2.56	-2.44	-2.32	-2.12	365	1.01
lpred[7]	-2.45	9.3e-3	0.18	-2.83	-2.56	-2.44	-2.32	-2.12	365	1.01
lpred[8]	-1.6	0.01	0.2	-2.04	-1.72	-1.59	-1.47	-1.24	376	1.0
lpred[9]	-2.45	9.3e-3	0.18	-2.83	-2.56	-2.44	-2.32	-2.12	365	1.01
lpred[10]	-2.49	9.8e-3	0.19	-2.87	-2.62	-2.48	-2.36	-2.15	367	1.01

Figure 1: Convergence Diagnostics Plots for the Fixed Effects Model

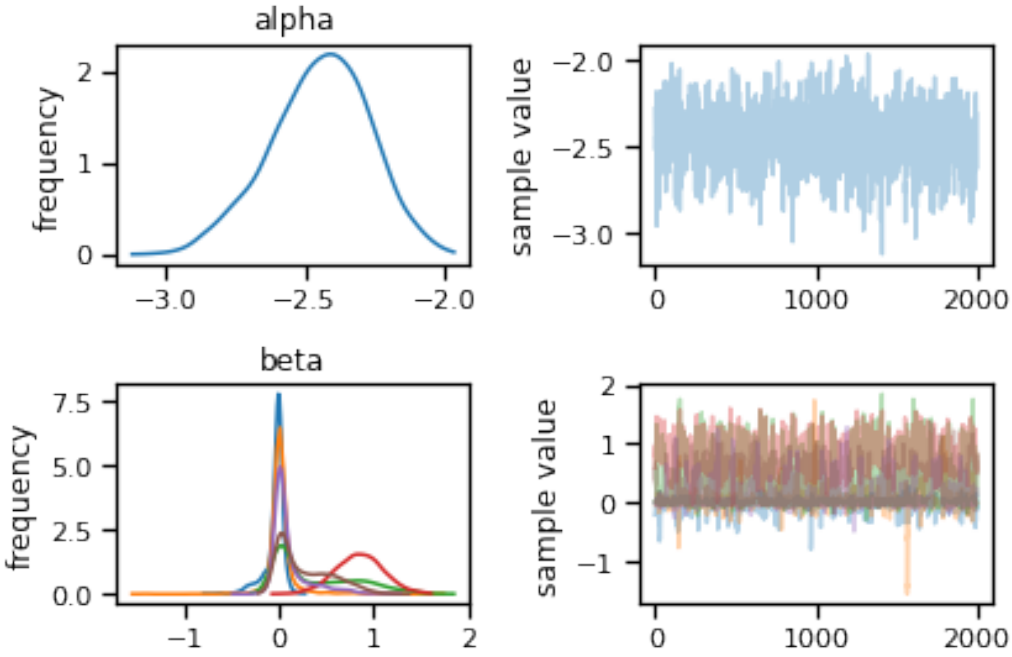
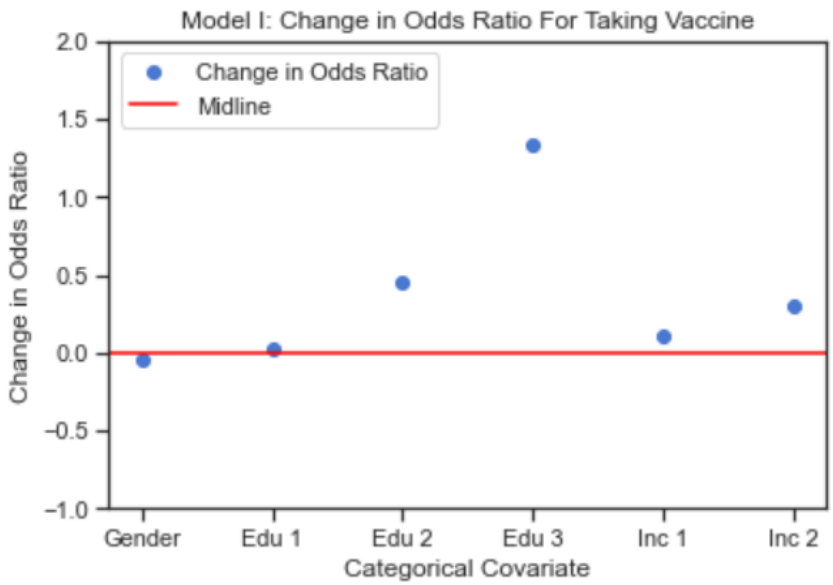


Figure 2: Model I: Plot of Change in Odds Ratios For Each Categorical Covariate



As regards the second model, the inferences are made individually for each region for all observations.[9] It can be seen from the figure 3(b) below that the random intercepts model has one regression line each for the eight regions. Technically, each of these lines is parallel to the overall intercept or average line under the fixed effects model in the first pooled fitted model. Hence, the difference, whether positive or negative, between each regional intercept, α_k, and the overall intercept α, gives the regional effect, α_{kj}, on the response. That is, $\alpha_{kj} = \alpha + \alpha_k$. For instance, the overall intercept from the single regression in model 1 is given as -2.45. Now, the intercept value of -1.95 for region 1 is a combination of the overall intercept value for the single regression line and the estimate of the random component pertaining to region 1, which value is 0.5. That is, $-1.95 = -2.45 + 0.5$. Put differently, the estimated intercepts for the eight regions indicate how much the overall intercept shifts up or down in particular regions. Thus, for example, in region 3 which is the Mid-West, the estimated intercept is -0.06 lower than the original average. And this implies that, people living in states within the Mid-West region of the U.S are less likely to agree with the mandatory vaccination policy, on average, holding all other things constant. In fact, the same is true for the West Coast, Mid-West, New England, Mid-Atlantic and Great Plains, which are represented by regions 2, 3, 6, 7 and 8 respectively. Starkly, the same conclusion remains for inferences made at the individual level within different geographical regions embodying a group or collection of states. The negative values pertaining to all y-hats under the posterior means column in table 2 below clearly show that, from a random intercepts point of view, the mandatory vaccination policy of the federal government is not totally acceptable by the general public. Included are the convergence diagnostics trace plots and posterior distributions plots in figure 1.

For the interpretation of slopes, the respective odds for taking a vaccine shot has a marginal increase for each of the categorical predictors holding all other factors constant. For instance, the odds for taking the vaccine by a male in region one rise marginally by 1%. Also, the odds rise by 84%, for for Educ level 3 and 6% for Inc level 1, holding all other factors constant. Similar to the dynamics on the change in odds for model I, model II experiences a fall in the odds for taking the vaccine by male gender. However, each of the covariates experiences a rise in the odds for taking the vaccine, with

Table 2: Output for the Varying Intercepts Model

```
Inference for Stan model: anon_model_a4d10c5790b1555b1b4d67f066b1356a.
4 chains, each with iter=1000; warmup=500; thin=1;
post-warmup draws per chain=500, total post-warmup draws=2000.
```

	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
alpha[1]	-1.95	0.01	0.25	-2.47	-2.13	-1.95	-1.78	-1.48	632	1.01
alpha[2]	-3.04	0.01	0.34	-3.78	-3.26	-3.04	-2.8	-2.4	669	1.01
alpha[3]	-2.51	0.01	0.32	-3.15	-2.71	-2.5	-2.28	-1.9	806	1.01
alpha[4]	-2.24	0.01	0.28	-2.8	-2.44	-2.24	-2.04	-1.7	714	1.01
alpha[5]	-2.13	0.01	0.32	-2.8	-2.35	-2.12	-1.91	-1.56	801	1.0
alpha[6]	-2.62	0.01	0.41	-3.49	-2.89	-2.59	-2.32	-1.88	1420	1.0
alpha[7]	-2.76	0.01	0.34	-3.47	-2.99	-2.73	-2.51	-2.12	934	1.0
alpha[8]	-2.82	0.02	0.5	-3.88	-3.14	-2.79	-2.49	-1.9	1104	1.0
beta[1]	-0.05	6.4e-3	0.11	-0.36	-0.07	-0.02	8.5e-3	0.08	300	1.01
beta[2]	0.03	6.0e-3	0.14	-0.15	-0.02	6.9e-3	0.04	0.38	518	1.01
beta[3]	0.37	0.03	0.42	-0.05	0.02	0.19	0.69	1.27	272	1.0
beta[4]	0.84	0.01	0.27	0.23	0.68	0.85	1.03	1.34	373	1.01
beta[5]	0.08	0.01	0.19	-0.1	-9.8e-3	0.02	0.09	0.69	265	1.02
beta[6]	0.21	0.01	0.26	-0.05	9.7e-3	0.09	0.37	0.85	336	1.01
lpred[1]	-2.18	0.01	0.32	-2.87	-2.39	-2.17	-1.96	-1.6	849	1.01
lpred[2]	-2.13	0.01	0.32	-2.8	-2.35	-2.12	-1.91	-1.56	801	1.0
lpred[3]	-2.56	0.01	0.32	-3.2	-2.75	-2.55	-2.33	-1.95	867	1.01
lpred[4]	-2.25	0.01	0.35	-3.0	-2.49	-2.24	-2.01	-1.6	824	1.0
lpred[5]	-3.09	0.01	0.35	-3.85	-3.31	-3.08	-2.86	-2.44	700	1.01
lpred[6]	-2.24	0.01	0.28	-2.8	-2.44	-2.24	-2.04	-1.7	714	1.01
lpred[7]	-2.13	0.01	0.32	-2.8	-2.35	-2.12	-1.91	-1.56	801	1.0
lpred[8]	-1.4	9.8e-3	0.31	-2.04	-1.6	-1.4	-1.19	-0.84	1000	1.0
lpred[9]	-2.24	0.01	0.28	-2.8	-2.44	-2.24	-2.04	-1.7	714	1.01
lpred[10]	-2.0	0.01	0.26	-2.54	-2.17	-1.99	-1.83	-1.52	645	1.01
lpred[11]	-1.5	0.02	0.39	-2.18	-1.78	-1.53	-1.23	-0.7	499	1.0
lpred[12]	-1.09	0.01	0.34	-1.77	-1.32	-1.08	-0.86	-0.44	882	1.0
lpred[13]	-2.13	0.02	0.44	-2.93	-2.44	-2.16	-1.83	-1.22	494	1.0
lpred[14]	-2.24	0.01	0.28	-2.8	-2.44	-2.24	-2.04	-1.7	714	1.01
lpred[15]	-3.09	0.01	0.35	-3.85	-3.31	-3.08	-2.86	-2.44	700	1.01

the highest rise occuring at Educ level 3 covariate, "holding all other factors constant. Meanwhile, the dynamics change and the conclusion differs for the same analysis in the remaining regions. Whereas the respective odds for taking the shot rise for each of the covariates in region 1, there is an alternating pattern for all other regions. In some regions, the odds for some of the covariates fall, while they rise for the remaining covariates. Taking region 2, for example, whereas the odds for taking the vaccine by gender male fall by 2%, and by 5%, for Educ level 1, they rise marginally by 1% for Inc level 3, "holding all other factors constant." A plot of the change in odds for each of the predictor variables is shown in figure 5.

Figure 3: Convergence Diagnostics Plots for the Varying Intercepts Model

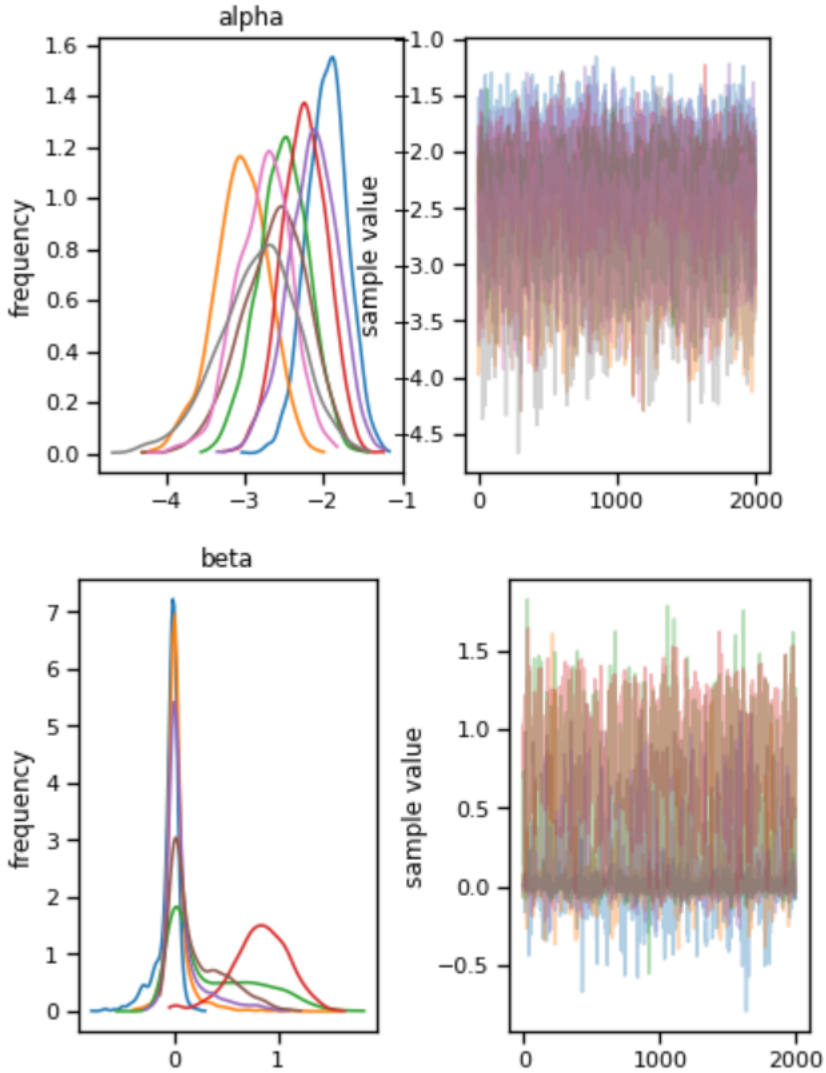


Figure 4: Model II: Plot of Regional Intercepts (alphas)

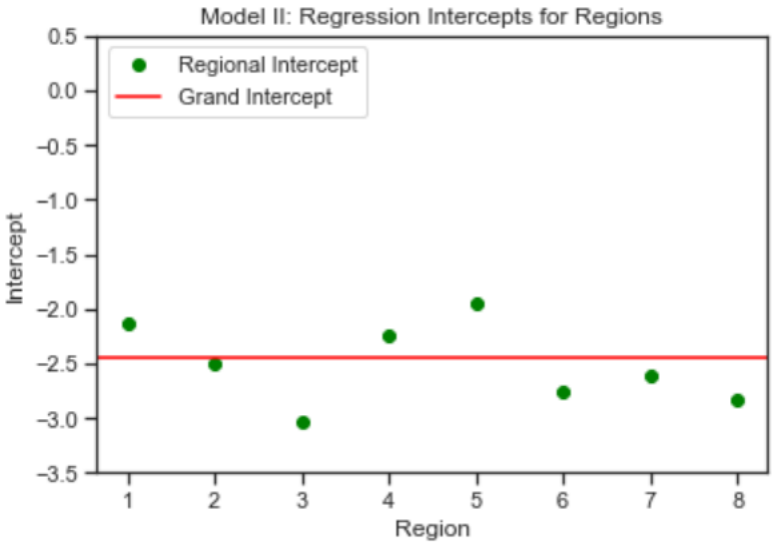


Figure 5: Model II: Plot of Odds Ratios for Each Categorical Covariate

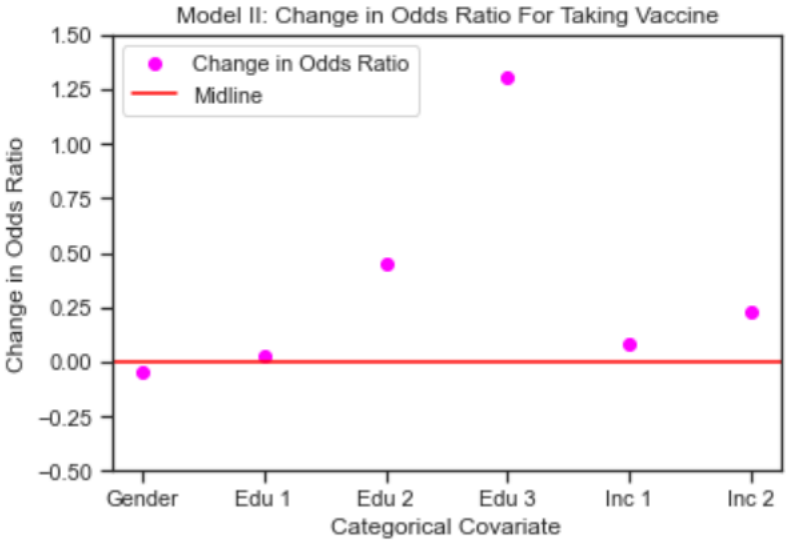
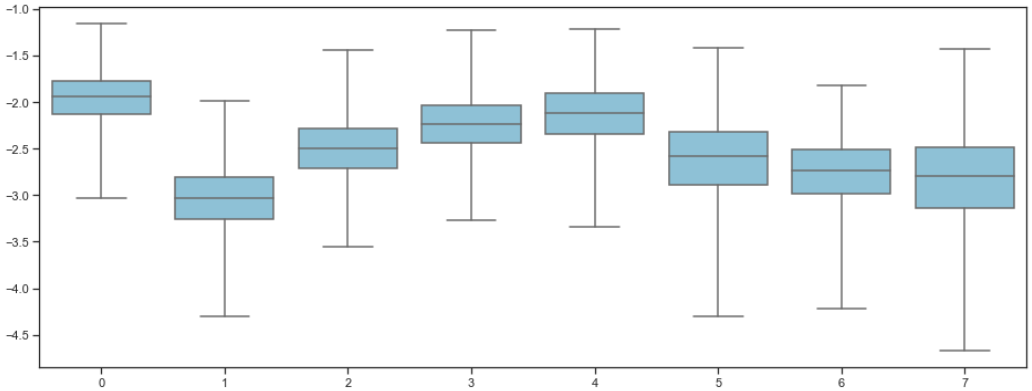


Figure 6: Parallel Box Plots for the Distributions of the Varying Intercepts Model



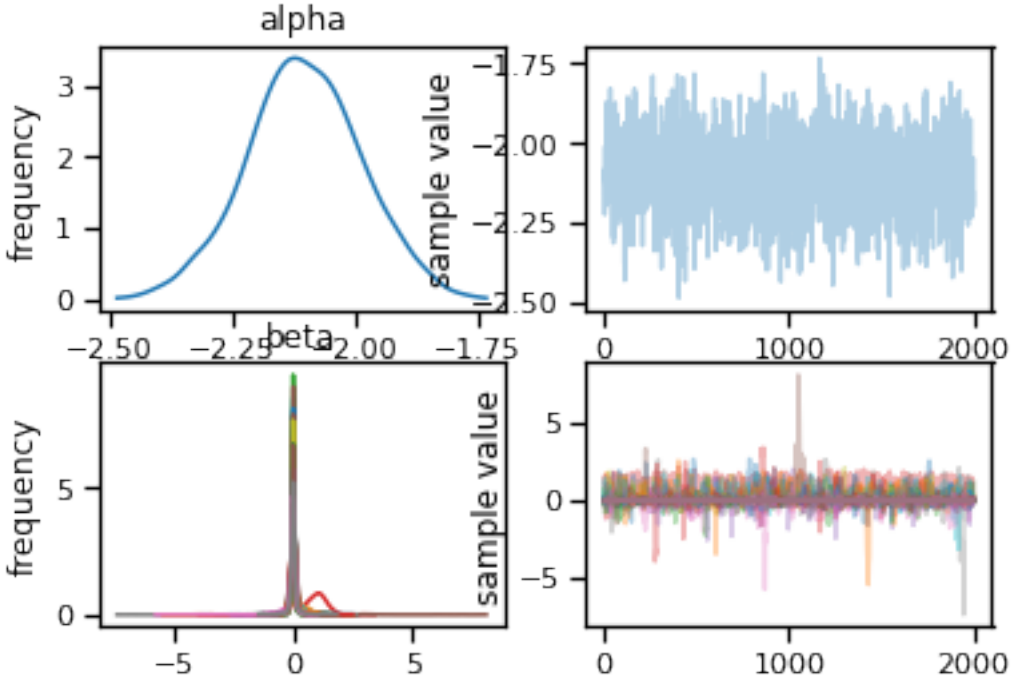
With regards to the final model, randomizing the variable coefficients on account of the regional level shows different model dynamics and conclusions[10] as can be seen from the output in table 3. In region one, the respective odds for taking a vaccine shot has a marginal increase for each of the categorical predictors holding all other factors constant. For instance, the odds for taking the shot by a male gender in region one rises by a marginal 1%. Similarly, the odds rise by 84% as a person downgrades from the $4th$ educational level to the $3rd$ level, and by 6% as an individual moves down from the $3rd$ income level to the $1st$ level, all in region one, holding the rest of the factors constant. However, the dynamics change and the conclusions differ for the same analysis in the remaining regions. Whereas the respective odds for taking the shot rises for each of the covariates in region one, there is an alternating pattern for all other regions. In some regions, the odds for some of the covariates fall, while they rise for the remaining covariates. For instance, taking region 2, for example, while the odds for taking the vaccine by gender male fall by 2% and by 5% as one moves from educational level 4 to 1, it rises by a marginal 1%, as the one moves from income level three to one, holding all other things constant.

Figure 7: Output for the Varying Slopes Model

```
Inference for Stan model: anon_model_445cd34050bdbb480d2695df89a45d34.
4 chains, each with iter=1000; warmup=500; thin=1;
post-warmup draws per chain=500, total post-warmup draws=2000.
```

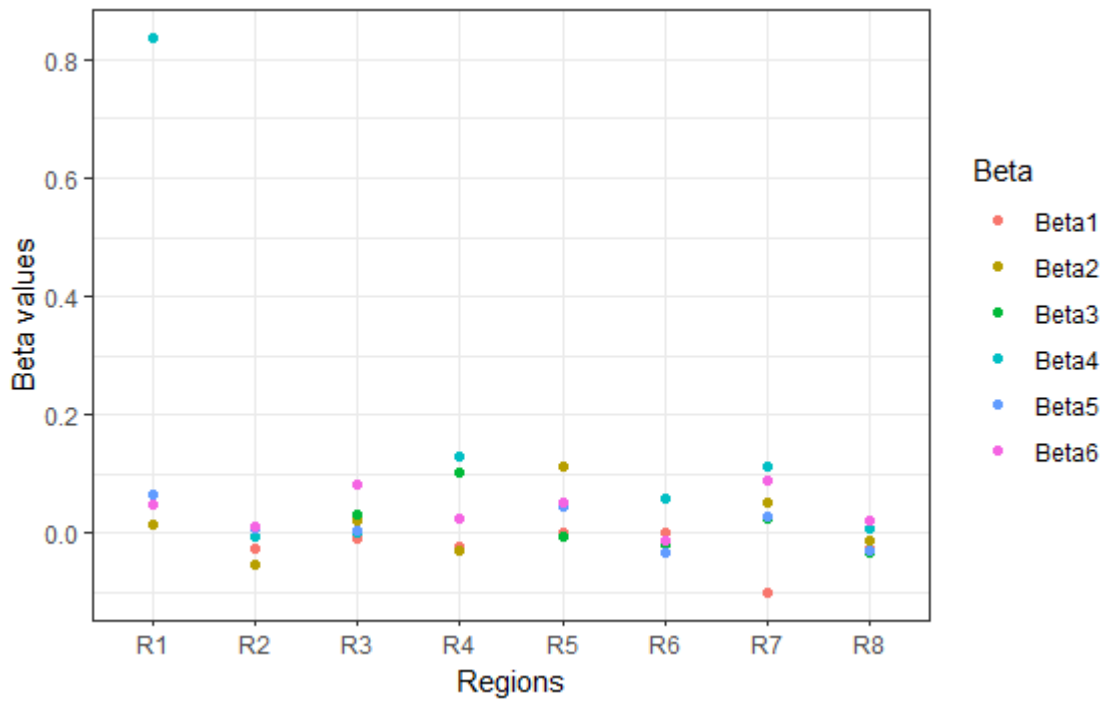
	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
alpha	-2.1	3.4e-3	0.12	-2.34	-2.18	-2.11	-2.03	-1.87	1167	1.0
beta[1,1]	0.01	3.2e-3	0.1	-0.15	-0.02	5.4e-3	0.04	0.27	947	1.0
beta[2,1]	0.02	9.2e-3	0.15	-0.22	-0.02	4.4e-3	0.04	0.34	268	1.01
beta[3,1]	0.05	9.1e-3	0.18	-0.16	-0.02	7.7e-3	0.05	0.65	403	1.02
beta[4,1]	0.84	0.03	0.5	-0.02	0.49	0.91	1.2	1.68	343	1.01
beta[5,1]	0.06	0.02	0.24	-0.12	-0.02	9.4e-3	0.05	0.78	230	1.01
beta[6,1]	0.05	0.01	0.19	-0.15	-0.02	6.7e-3	0.05	0.55	290	1.01
beta[1,2]	-0.02	5.8e-3	0.13	-0.38	-0.03	-4.5e-3	0.02	0.17	540	1.01
beta[2,2]	-0.05	0.05	0.45	-0.69	-0.03	-1.6e-3	0.03	0.33	95	1.04
beta[3,2]	9.4e-3	4.1e-3	0.11	-0.16	-0.03	8.9e-4	0.03	0.24	772	1.01
beta[4,2]	-6.5e-3	3.3e-3	0.1	-0.22	-0.03	-2.0e-3	0.02	0.17	873	1.0
beta[5,2]	7.0e-3	4.3e-3	0.14	-0.25	-0.03	1.2e-3	0.03	0.32	1073	1.0
beta[6,2]	0.01	3.9e-3	0.12	-0.22	-0.02	1.9e-3	0.03	0.3	953	1.0
beta[1,3]	-7.3e-3	2.0e-3	0.08	-0.2	-0.03	-1.3e-3	0.02	0.14	1825	1.0
beta[2,3]	0.02	0.02	0.33	-0.28	-0.03	8.9e-4	0.03	0.36	180	1.02
beta[3,3]	0.03	8.4e-3	0.17	-0.17	-0.02	4.1e-3	0.04	0.57	396	1.01
beta[4,3]	9.5e-4	2.7e-3	0.09	-0.2	-0.03	7.0e-4	0.03	0.2	1062	1.0
beta[5,3]	3.5e-3	4.4e-3	0.14	-0.26	-0.03	5.6e-4	0.03	0.3	1076	1.0
beta[6,3]	0.08	0.01	0.24	-0.13	-0.02	0.01	0.07	0.92	409	1.01
beta[1,4]	-0.02	5.3e-3	0.11	-0.35	-0.04	-4.6e-3	0.02	0.16	457	1.01
beta[2,4]	-0.03	0.03	0.26	-0.54	-0.03	-1.8e-3	0.03	0.26	109	1.04
beta[3,4]	0.1	0.03	0.36	-0.14	-0.02	0.01	0.05	1.31	197	1.01

Figure 8: Convergence Diagnostics Plots for the Varying Slopes Model



While we diagnose the third model, we intend to investigate the necessity of fitting such a model in which all coefficients for the six covariates vary individually for each of the eight regions. Thus, a final plot figure is presented that pictorially presents the scenario. In the meantime, a formal hypothesis testing could be performed to either uphold or debunk this claim, but the results from the plot in figure 8 do send in a signal that for each region, the regression coefficients do differ from each other, although some of the differences are not so significant. This is shown in figure 8.

Figure 9: Model III: Plot of Each of Six Regression Coefficients for all Eight Regions



4.1 Analysis of Variance Tests for Model II

Hypothesis for Equal Regional Intercepts

Hypothesis :

$H_0 : \alpha_1 = \alpha_2 = ... = \alpha_8 = 0$

H_1 : at least one $\alpha_i \neq 0$ for i $\neq j$

Test Statistic :
$F^* = \frac{MSR}{MSE} \sim F^{0.05}_{7,15992}$
Decision Rule :
Reject H_0 if $F^* > F^{0.05}_{7,15992}$. Otherwise, fail to reject H_0.

Figure 10: Model II: Analysis of Variance Test for Regression Intercepts

	sum_sq	df	F	PR(>F)
C(regions)	1955.328781	7.0	2230.721684	0.0
Residual	2002.530526	15992.0	NaN	NaN

Decision :
Since $F^* = 2230.72 > F^{0.05}_{7,15992} = 2.51$ in in figure 9, we reject H_0.
Conclusion :
At a 5% level of significance, we can conclude that there exist significant differences among the population intercept means for the eight different regions.

Post-Hoc Tukey HSD Multiple Pairwise Comparisons Test
Hypothesis :
H_0 : Each pair $\alpha_i = \alpha_j$ for i $\neq j$
H_1 : None of the pairs equal, for i $= 1, .., 8$ and i $\neq j$
Test Statistic :
$q_s = \frac{\bar{y}_{\max} - \bar{y}_{\min}}{\sqrt{\frac{MSE}{n}}} \sim$ Studentized Range Distribution, where: q_s is the calculated studentized range value; $\bar{y}_{\max}$ is the largest of the treatment means, and $\bar{y}_{\min}$ is the smallest of treatment means. Moreover, the q_s value can be compared to a q value from the studentized range distribution.
Decision Rule :
Reject H_0 if p-vlue is less than significance level. Otherwise, fail to reject H_0.. Alternatively, If the q_s value is larger than the critical value q_α obtained from the distribution, the two means are said to be significantly different at level $\alpha : 0 < \alpha < 1$.

Decision :

Figure 11: Model II: Tukey's HSD Test for Multiple Pairwise Comparisons

Multiple Comparison of Means - Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
R1	R2	-1.0871	0.001	-1.121	-1.0532	True
R1	R3	-0.5538	0.001	-0.5877	-0.5199	True
R1	R4	-0.2857	0.001	-0.3196	-0.2517	True
R1	R5	-0.1797	0.001	-0.2136	-0.1458	True
R1	R6	-0.6618	0.001	-0.6957	-0.6279	True
R1	R7	-0.8035	0.001	-0.8375	-0.7696	True
R1	R8	-0.8712	0.001	-0.9051	-0.8373	True
R2	R3	0.5333	0.001	0.4993	0.5672	True
R2	R4	0.8014	0.001	0.7675	0.8353	True
R2	R5	0.9074	0.001	0.8735	0.9413	True
R2	R6	0.4253	0.001	0.3914	0.4592	True
R2	R7	0.2835	0.001	0.2496	0.3175	True
R2	R8	0.2159	0.001	0.182	0.2498	True
R3	R4	0.2682	0.001	0.2342	0.3021	True
R3	R5	0.3741	0.001	0.3402	0.408	True
R3	R6	-0.108	0.001	-0.1419	-0.074	True
R3	R7	-0.2497	0.001	-0.2836	-0.2158	True
R3	R8	-0.3174	0.001	-0.3513	-0.2834	True
R4	R5	0.106	0.001	0.072	0.1399	True
R4	R6	-0.3761	0.001	-0.41	-0.3422	True
R4	R7	-0.5179	0.001	-0.5518	-0.484	True
R4	R8	-0.5855	0.001	-0.6194	-0.5516	True
R5	R6	-0.4821	0.001	-0.516	-0.4482	True
R5	R7	-0.6238	0.001	-0.6578	-0.5899	True
R5	R8	-0.6915	0.001	-0.7254	-0.6576	True
R6	R7	-0.1418	0.001	-0.1757	-0.1078	True
R6	R8	-0.2094	0.001	-0.2433	-0.1755	True

From the tabulated results of the Tukey's HSD post anova analysis in figure 10, it could be observed the p-values of the comparison between all possible pairwise combinations (i.e. $\binom{8}{2} = 28$) of the mean difference between regional intecepts are all equal to 0.001. And since this is less than the significance level of 0.05, the null hypothesis of zero pairwise mean-difference for any pair is rejected at 5%.

<u>Conclusion</u> :

Thus, we conclude that there is a 5% statistically significant difference between the means of any pair of the regional intercepts, for the eight different regions.

Similar results are gleaned from output obtained from Scheffe's and Dunnett's Bonferroni $p-value$ adjusted multiple pairwise comparisons test. Since the two tables are similar, only Scheffe's test results output is included in this report in figure 12 below. It can be concluded from the results in the table below that each pair is significantly different at a 5%.

Figure 12: Model II: Scheffe's Post Hoc Multiple Pairwise Comparisons Test

```
sp.posthoc_scheffe(new_sample, val_col = 'value', group_col = 'regions')
```

	R1	R2	R3	R4	R5	R6	R7	R8
R1	1.000000e+00	0.000000e+00	0.000000e+00	1.033210e-133	1.569253e-51	0.000000e+00	0.000000e+00	0.000000e+00
R2	0.000000e+00	1.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	1.769813e-294	1.063084e-131	1.735370e-75
R3	0.000000e+00	0.000000e+00	1.000000e+00	1.211281e-117	4.596630e-229	3.248059e-17	8.910730e-102	3.371219e-165
R4	1.033210e-133	0.000000e+00	1.211281e-117	1.000000e+00	1.610146e-16	1.788934e-231	0.000000e+00	0.000000e+00
R5	1.569253e-51	0.000000e+00	4.596630e-229	1.610146e-16	1.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
R6	0.000000e+00	1.769813e-294	3.248059e-17	1.788934e-231	0.000000e+00	1.000000e+00	3.698738e-31	7.063297e-71
R7	0.000000e+00	1.063084e-131	8.910730e-102	0.000000e+00	0.000000e+00	3.698738e-31	1.000000e+00	5.815801e-06
R8	0.000000e+00	1.735370e-75	3.371219e-165	0.000000e+00	0.000000e+00	7.063297e-71	5.815801e-06	1.000000e+00

5 An Ad-hoc Model to Consider

5.0.1 Model IV: A Varying Regional Effect and Varying Covariate Effect within a Region

Consequent to considering the previous three models, a final model that investigates the existence of a varying intercept for each individual within a region and a varying slope across all regions is being considered. The baseline for a region is given as α_k. Thus, for the $K = 8$ regions:

$$y_{kj} \sim \text{Bernoulli}(\pi_{kj})$$

The model at region k becomes:

$$\mathbb{P}(y_{kj} = 1) = \pi_{kj} = \frac{e^{\alpha_k + \boldsymbol{x}'_{kj}\beta_k}}{1 + e^{\alpha_k + \boldsymbol{x}'_{kj}\beta_k}}$$

where: $k = 1, 2, ..., K$; $j = 1, 2, ..., n_k$; $\alpha = (\alpha_1, \alpha_2, ..., \alpha_k)'$; $\beta_k = (\beta_{k1}, \beta_{k2}, ..., \beta_{k6})'$; $\boldsymbol{x}_{kj} = (x_{kj1}, ..., x_{kj6})'$; and n_k = total number of observations or units within the k-th region. Both α and β vary from region to region.

5.1 Output Results For the Fourth Model

Table 3: Output for the Random Slopes and Intercepts (Model IV)

	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
alpha[1]	-1.5	9.4e-3	0.26	-2.08	-1.65	-1.48	-1.32	-1.04	787	1.0
alpha[2]	-2.49	8.6e-3	0.32	-3.19	-2.69	-2.47	-2.27	-1.91	1382	1.0
alpha[3]	-2.09	0.01	0.34	-2.84	-2.3	-2.07	-1.87	-1.49	841	1.0
alpha[4]	-1.77	7.9e-3	0.3	-2.39	-1.96	-1.76	-1.56	-1.21	1440	1.0
alpha[5]	-1.82	8.8e-3	0.35	-2.54	-2.05	-1.82	-1.59	-1.16	1624	1.0
alpha[6]	-2.22	0.01	0.42	-3.16	-2.46	-2.2	-1.93	-1.47	1610	1.0
alpha[7]	-2.54	0.02	0.5	-3.65	-2.82	-2.48	-2.18	-1.74	423	1.0
alpha[8]	-2.16	0.01	0.51	-3.27	-2.48	-2.12	-1.81	-1.26	1708	1.0
beta[1,1]	-2.1e-3	2.9e-3	0.11	-0.24	-0.03	-1.5e-4	0.03	0.22	1342	1.0
beta[2,1]	3.1e-3	1.9e-3	0.08	-0.15	-0.02	1.6e-3	0.03	0.17	1752	1.0
beta[3,1]	0.06	0.02	0.22	-0.15	-0.02	6.9e-3	0.04	0.79	143	1.02
beta[4,1]	0.12	0.02	0.29	-0.1	-0.01	0.02	0.09	1.05	259	1.01
beta[5,1]	0.04	0.01	0.19	-0.17	-0.02	6.0e-3	0.04	0.53	300	1.01
beta[6,1]	0.02	6.7e-3	0.14	-0.18	-0.02	2.3e-3	0.03	0.34	431	1.01
beta[1,2]	0.03	5.8e-3	0.16	-0.16	-0.02	5.0e-3	0.04	0.4	757	1.01
beta[2,2]	-0.19	0.2	1.58	-0.57	-0.03	-8.2e-4	0.03	0.21	63	1.04
beta[3,2]	0.01	5.8e-3	0.15	-0.25	-0.03	3.3e-3	0.03	0.4	679	1.0
beta[4,2]	0.03	7.9e-3	0.15	-0.13	-0.02	4.7e-3	0.04	0.47	368	1.03

Figure 13: Convergence Diagnostics Plots Model IV

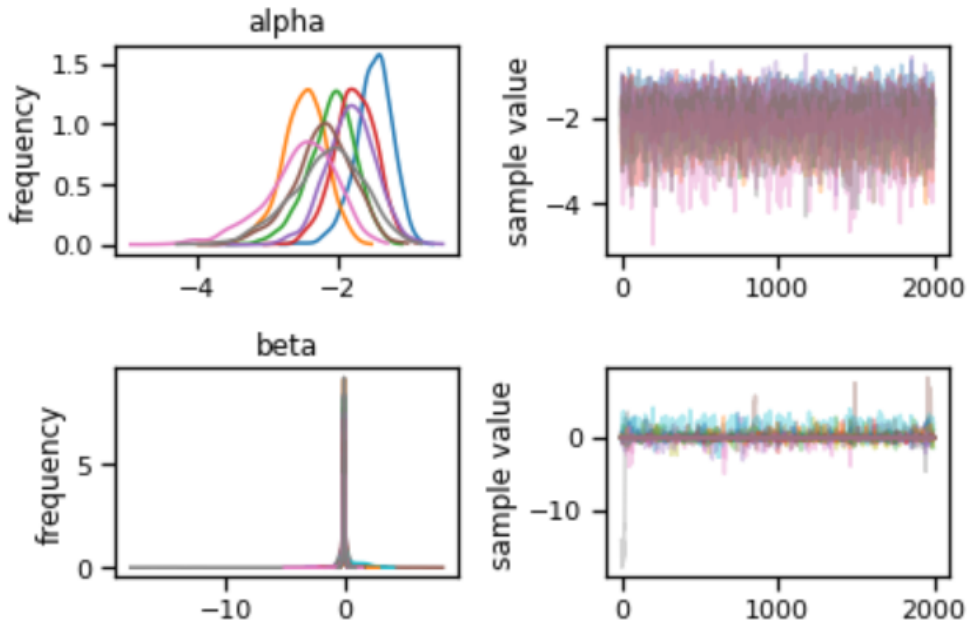
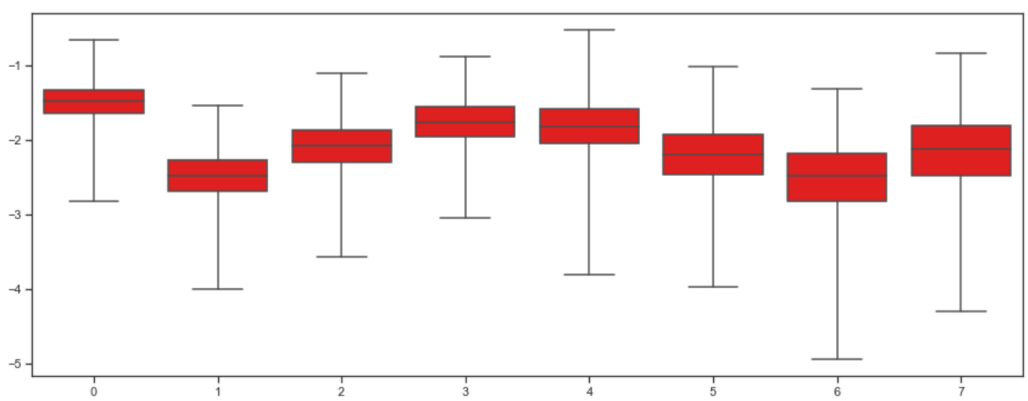


Figure 14: Parallel Box Plots for the Distributions of the Varying Intercepts (Model IV)



5.2 Analysis of Variance Tests for Model IV

Hypothesis for Equal Regional Intercepts

Hypothesis :

$H_0 : \alpha_1 = \alpha_2 = ... = \alpha_8 = 0$

H_1 : at least one $\alpha_i \neq 0$ for i $\neq j$
Test Statistic :
$F^* = \frac{MSR}{MSE} \sim F^{0.05}_{7,15992}$
Decision Rule :
Reject H_0 if $F^* > F^{0.05}_{7,15992}$. Otherwise, fail to reject H_0.

Figure 15: Model IV: Analysis of Variance Test for Regression Intercepts

	sum_sq	df	F	PR(>F)
C(regions)	1815.701471	7.0	1742.66049	0.0
Residual	2380.325788	15992.0	NaN	NaN

Decision :
Since $F^* = 2230.72 > F^{0.05}_{7,15992} = 2.51$ in figure 15, we reject H_0.
Conclusion :
At a 5% level of significance, we can conclude that there exist significant differences among the population intercept means for the eight different regions.

Post-Hoc Tukey HSD Multiple Pairwise Comparisons Test
Hypothesis :
H_0 : Each pair $\alpha_i = \alpha_j$ for i $\neq j$
H_1 : None of the pairs equal, for i $= 1, .., 8$ and i $\neq j$
Test Statistic :
$q_s = \frac{\bar{y}_{\max} - \bar{y}_{\min}}{\sqrt{\frac{MSE}{n}}} \sim$ Studentized Range Distribution, where: q_s is the calculated studentized range value; $\bar{y}_{\max}$ is the largest of the treatment means, and $\bar{y}_{\min}$ is the smallest of treatment means. Moreover, the q_s value can be compared to a q value from the studentized range distribution.
Decision Rule :
Reject H_0 if p-vlue is less than significance level. Otherwise, fail to reject H_0.. Alternatively, If the q_s value is larger than the critical value q_α obtained from the distribution, the two means are said to be significantly different at level $\alpha : 0 < \alpha < 1$.

Decision :

Figure 16: Model IV: Tukey's HSD Test for Multiple Pairwise Comparisons

Multiple Comparison of Means - Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
R1	R2	-0.9934	0.001	-1.0304	-0.9564	True
R1	R3	-0.5942	0.001	-0.6312	-0.5573	True
R1	R4	-0.2668	0.001	-0.3038	-0.2298	True
R1	R5	-0.3224	0.001	-0.3594	-0.2855	True
R1	R6	-0.7211	0.001	-0.7581	-0.6841	True
R1	R7	-1.0371	0.001	-1.0741	-1.0001	True
R1	R8	-0.6652	0.001	-0.7021	-0.6282	True
R2	R3	0.3992	0.001	0.3622	0.4362	True
R2	R4	0.7266	0.001	0.6896	0.7636	True
R2	R5	0.671	0.001	0.634	0.708	True
R2	R6	0.2723	0.001	0.2353	0.3093	True
R2	R7	-0.0437	0.0082	-0.0807	-0.0067	True
R2	R8	0.3283	0.001	0.2913	0.3652	True
R3	R4	0.3274	0.001	0.2905	0.3644	True
R3	R5	0.2718	0.001	0.2348	0.3088	True
R3	R6	-0.1269	0.001	-0.1638	-0.0899	True
R3	R7	-0.4429	0.001	-0.4798	-0.4059	True
R3	R8	-0.0709	0.001	-0.1079	-0.0339	True
R4	R5	-0.0556	0.001	-0.0926	-0.0187	True
R4	R6	-0.4543	0.001	-0.4913	-0.4173	True
R4	R7	-0.7703	0.001	-0.8073	-0.7333	True
R4	R8	-0.3984	0.001	-0.4353	-0.3614	True
R5	R6	-0.3987	0.001	-0.4356	-0.3617	True
R5	R7	-0.7147	0.001	-0.7516	-0.6777	True

From the tabulated results of the Tukey's HSD post anova analysis in figure 16, it could be observed the p-values of the comparison between all possible pairwise combinations (i.e. $\binom{8}{2} = 28$) of the mean difference between regional intecepts are all equal to 0.001. And since this is less than the significance level of 0.05, the null hypothesis of zero pairwise mean-difference for any pair is rejected at 5%.

Conclusion :

Thus, we conclude that there is a 5% statistically significant difference between the means of any pair of the regional intercepts, for the eight different regions.

5.3 Prediction

The fourth model was used to make predictions on a test data and it was found that it made a fair prediction accuracy of 86%. Better still, a prediction mean square error of 14% was realized when the model was used to make prediction based on a test data.

6 Summary and Conclusion

The work began by considering a pooled fitted regression model case that assumes no inter-regional or inter-state variations among all sampling units[10]. It went ahead to look at a second case that recognizes and considers possible variations among the eight regional groupings of the United States, and thus fits a varying intercepts model that randomizes the model intercepts based on the eight regional groupings[4]. Finally, for model comparison purposes, a third case considers a varying slopes model that randomizes the model coefficients on the basis of the regional groupings.[11] It was realized from the above results that there is an overall repulsion by the general public to the mandatory covid-19 policy by the federal government of the United States, and one could conclude that, there must be other factors, rather than the selected socioeconomic factors of gender, income and education above, influencing the implementation of the policy, however profitable it must be. While various articles have cited unfounded health-related causes, others have pointed to shrouded political undertones as the main driving forces for the mandatory implementation of the vaccination policy. Whichever way it may go, we recommend detailed research for investigation into these matters.

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